The Missing Coordinate Theory:
Event Priority, Non-Universal Measurability, and Ontological Inversion Across Physics, AI, Institutions, and Lived Experience

0. ABSTRACT

One paragraph, 200–300 words.
Include:

- Primitive events $\mathcal{E}$ vs. coordinate events $C_i(\mathcal{E})$
- **Ontological inversion:** representation governs the event
- **Axiom 6:** non-universal measurability
- **Proposition 8:** missing dimensions
- **Cross-domain claim:** same structure appears in physics, AI, and institutions
- **One sentence on the RQM intervention:** your direct event definition and actualization solution
- **One sentence on AI:** persistent corrected memory as the missing coordinate
- **One sentence on institutions:** four filters produce bounded correctness
- **One sentence on the formal contribution:** axioms, propositions, operators, loss functional, fiber-bundle geometry

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The Missing Coordinate Theory claims that:

1. **Events are ontologically primary.**
   What truly exists are primitive events: concrete changes in the world with all their consequences—physical, biological, psychological, social, historical.
2. **Dimensions and coordinates are secondary tools.**
   Space, time, metrics, variables, legal categories, diagnostic codes, quantum states, economic indicators, etc., are coordinate systems invented to index and compare events, not the events themselves.
3. **Modern civilization has inverted this order.**
   Systems treat what can be measured, coded and authorized as *the whole of reality*, while everything that falls outside these coordinates is treated as if it were ontologically negligible, even when it is experientially decisive.
4. **As a result, we live under a "regime of bounded correctness".**
   Science, AI, institutions and governance can be very correct inside their chosen coordinates, yet systematically fail the actual events they are

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supposed to deal with. The problem is not that they are “wrong”, but that they are structurally incomplete.

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1. Introduction

1.1 Motivation and Central Question

Why do all of the following display the same structural pathology?

- quantum measurement and QM–GR unification
- black-hole horizons and cosmological singularities
- AI hallucination, statelessness, and non-learning
- institutional failures in medicine, law, and governance
- chronic illness and lived-system failure

State the core hypothesis: A single ontological error, treating coordinate representations as reality itself,is being repeated across domains.

1.2 Primitive Events vs. Coordinate Events (First Intuitive Pass)

Define:

Primitive event: $\mathcal{E}$**:** full transformed reality

- Who it happened to
- In what context and power relations
- With what chain of physical, biological, emotional, relational, and social consequences

Not reducible to one metric or one time-space point

Coordinate event: $C_i(\mathcal{E})$: indexed representation inside a system

- A spacetime point ((t, x, y, z))
- A quantum state vector or measurement outcome
- A medical ICD code
- A legal charge in a case file
- Database entry
- KPI / system log

Key Claim:

Coordinates are indexing tools, not reality itself. The fundamental error of modern systems is to treat the index as the thing.

Central Claim of the Missing Coordinate Theory

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Primitive events are ontologically primary; dimensions and coordinates are secondary tools.

- **A primitive event $\mathcal{E}$:** the full, transformed reality of "what happened" – physical, biological, relational, historical.
- **A coordinate event $C_i(\mathcal{E})$:** the standardized representation inside some grid (spacetime point, quantum state, legal code, diagnosis, KPI, database row, etc.)

1.3 The 9/11 Example

- September 11, 2001, 6:00 AM, WTC, NYC
- September 11, 2001, 6:00 PM, WTC, NYC

Show: The coordinates register that something changed, but they do not contain smoke, collapse, toxic air, trauma, death, political rupture, or historical reality

Conclusion: Coordinates measure the delta; they do not carry the transformed reality of the event.

1.4 Ontological Inversion

Modern systems have undergone a civilizational inversion:

Measurement and coordinates were originally created to serve events, but have been elevated into governing reality. Systems no longer govern events directly; they govern authorized reconstructions.

Original order:

$\text{event} \rightarrow \text{measurement/representation}$

Modern inversion:

$\text{representation} \rightarrow \text{what counts as the event}$

This leads to a regime of bounded correctness:

- Systems can be *procedurally perfect, internally valid, statistically supported* and still be false to the actual events they are meant to govern, because the primitive event never acquires governing authority over its coordinate representation.

1.5 Why the Theory Says "Incomplete," Not "Incorrect"

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- Not attacking science or mathematics
- Identifying structural incompleteness
- The correction: representation must serve the event, not replace it

1.6 What the Theory Is NOT

- Not anti-science
- Not anti-measurement
- Not anti-formalization
- Not replacing rigor with subjectivity
- Not saying every lived account overrides every formal system
- Claiming instead that scientific measurability is powerful but incomplete as a total ontology for events
- The correction is that representation must serve the event, not replace it

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1.7 Scope and Structure of the Document

State that the paper moves from:

conceptual framing → formal structure → operators → physics → AI → institutions → lived cases → quantification → final statement

- Formulate an event-first ontology
- Formalize the axioms and propositions
- Insert the operators and functionals already derived
- Apply the theory to physics, AI, and institutions
- Show the scenario corpus as empirical illustration
- Close with the final formal statement and self-diagnosis

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2. Formal Vocabulary and Notation

This section defines the formal symbols used throughout the theory and specifies not only what each symbol denotes, but what role it plays in the event-first framework.

2.1 Core Symbols

$\mathcal{E}$ **— Primitive event.**

Use: the full event in its transformed reality, not just its measurable description.

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Ω **— Space or class of primitive events.**
Use: the domain from which primitive events are drawn.

O_i **— Observer or system i.**
Use: marks that every representation is relative to some observer, institution, apparatus, model, or system.

S **— Target system or interacting system.**
Use: the thing an observer interacts with in event actualization and relational physics examples.

$C_i : \Omega \to M_i$ **— Observer-relative coordinate map.**
Use: maps a primitive event into the representation used by observer/system i. This is the formal expression of how systems index events.

M_i **— Coordinate space for observer/system i.**
Use: the admissible representational space — spacetime coordinates, Hilbert space, medical coding system, legal file structure, database schema, or institutional category set.

$D_{current}$ **— Current dimensional framework.**
Use: the dimensions and variables a system already recognizes and treats as measurable or admissible.

$D^{(i)}_{missing}$ **— Missing dimensions relative to observer/system i.**
Use: aspects of the event excluded from M_i — lived transformation, hidden cost, stakes, relational context, historical consequence.

I_{ij} **— Interaction operator between systems i and j.**
Use: formalizes how two partial representations meet and produce a new joint representation.

$\Phi(\mathcal{E}, O_i)$ **— Actualization functional.**
Use: measures how strongly event $\mathcal{E}$ becomes real or governing for observer/system O_i. Distinguishes "not yet actualized," "actualized but not recognized," and "actualized and governing."

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$\Phi_{thr}(i)$ **— Actualization threshold for observer/system i.**
Use: the threshold above which an event counts as actualized for that observer/system. Can vary by context, architecture, or filter structure.

$M_t^{(i)}$ **— Memory state of AI/system i at time t.**
Use: tracks whether corrections persist or vanish across time.

$\alpha(i)$ **— Persistence or learning rate for system i.**
Use: measures how much a grounding event actually changes future system behavior.
- $\alpha = 0$: stateless predictor
- $\alpha > 0$: persistent correction is possible

$\hat{C}(\mathcal{E}_{ground}^{(t)})$ **— Encoded grounding event at time t.**
Use: the system-readable form of a real correction, consequence, or grounding interaction.

$L_i(\mathcal{E})$ **or** $\mathcal{L}_i(\mathcal{E})$ **— Governing-authority loss functional.**
Use: measures how badly observer/system i fails the primitive event. Combines representational mismatch with stakes.

F_{ont} **— Ontological filter.**
Use: determines what kinds of things are allowed to count as real in the system.

F_{auth} **— Authority filter.**
Use: determines who is allowed to introduce reality in governing form.

I_{int} **— Interpretive filter.**
Use: transforms incoming reality into the language and categories the system can process.
K_i **— Correctness predicate for system i.**
Use: defines what counts as "correct" inside that system's admissible frame.

$A(\mathcal{E})$ **— Acknowledged event.**
Use: indicates the event has been named, recorded, or recognized in some way.

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$G(\mathcal{E})$ **— Governing event.**
Use: indicates the event actually influences outcomes, decisions, or system behavior.

(E_i, B_i, π_i, F_i) **— Fiber-bundle objects.**
Use: geometric formulation of the theory.

- B_i: base space of coordinate events
- F_i: fiber of missing dimensions attached to each coordinate event
- E_i: total space combining both
- π_i: projection from total space to coordinate base

R_i **— Reconstruction operator for observer/system i.**
Use: describes how observer i rebuilds or remembers the event from available representation.

p_i^{prior}, p_i^{post} **— Prior and posterior descriptive distributions for observer/system i.**
Use: used in the actualization functional to measure change before and after interaction.

2.2 How These Symbols Function in the Theory

Event ontology: $\mathcal{E}$, Ω
Define what reality is fundamentally made of in the theory.

Representation: O_i, C_i, M_i, $D_{current}$, $D_{missing}^{(i)}$
Define how systems represent events and what gets left out.

Actualization and relation: S, I_{ij}, Φ, Φ_{thr}, R_i
Define when an event becomes real for a system and how different frames align.

Memory and AI: $M_t^{(i)}$, $\alpha(i)$, $\hat{C}(\mathcal{E}_{ground}^{(t)})$
Define whether a system can accumulate grounded correction across time.

Filters and bounded correctness: F_{ont}, F_{auth}, I_{int}, K_i, $A(\mathcal{E})$, $G(\mathcal{E})$, $L_i(\mathcal{E})$
Define how systems decide what counts, who counts, how reality is translated, what correctness means, and how badly an event is failed.

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Geometric upgrade: (E_i, B_i, π_i, F_i)
Let missing dimensions be expressed in fiber-bundle language.

2.3 GR Notation Separation

Reserve standard GR notation for spacetime structure:
- $p \in \mathcal{M}$: spacetime point
- $x^\mu(p)$: coordinates of p
- $g_{\mu\nu}$: metric
- τ: proper time

State explicitly: $\mathcal{E}$ is not identical to a GR spacetime point. A spacetime point may index part of an event, but does not exhaust the primitive event.

2.4 Presentation Rule

Each formal item follows:
name → verbal statement → equation → variable definitions → interpretation → domain use

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3. Foundational Definitions

3.1 Primitive Event

A primitive event $\mathcal{E}$ is the full, lived, transformed occurrence:
What happened, to whom, in what web of relations, with what stakes and consequences

- physical change
- experiential change
- relational consequences
- historical meaning
- downstream effects
- power context
- memory continuity or rupture
- long term consequences

Events are not just spacetime points

3.2 Coordinate / Dimension

Dimensions are secondary tools for comparing and indexing events.
They track:

- Time intervals
- Spatial displacement
- Quantitative deltas
- Other admissible measurable changes

They are tools for comparison not the event itself

3.3 Primitive Event vs. Coordinate Event (Formal)

$C_i : \Omega \to M_i$

Core dimensional inequality: $\dim(\mathcal{E}) > \dim(C_i(\mathcal{E})) \quad \forall i$
Interpretation: Events are strictly higher-dimensional than any of their coordinate projections. The map $\mathcal{E} \mapsto C_i(\mathcal{E})$ is necessarily many-to-one and lossy.

3.4 Ontological Inversion (Formal)

$\text{Inversion}(i) := C_i(\mathcal{E}) \succ \mathcal{E}$

where "$\succ$" means "is treated as having governing authority over."
Interpretation: the coordinate representation is elevated above the event it was created to serve.

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4. Axiomatic Framework

Each axiom follows: Name → Verbal statement → Formal statement → Interpretation

4.1 Axiom 1 — Ontological Priority of Events

Verbal: Events are ontologically prior to dimensions. No dimension precedes or constitutes the event it indexes.

Formal Statement: $\forall D_{current} : D_{current} \prec \mathcal{E} \;\Rightarrow\; \dim(\mathcal{E}) > \dim(D_{current})$

Interpretation: Events exist first; representations are derived.

4.2 Axiom 2 — Non-Exhaustibility of Representation

Verbal: No coordinate or dimensional description exhausts the event it represents.

Formal Statement: $\forall O_i,\ \forall \mathcal{E} : C_i(\mathcal{E}) \neq \mathcal{E} \;\wedge\; \dim(C_i(\mathcal{E})) < \dim(\mathcal{E})$

Interpretation: The map $\mathcal{E} \mapsto C_i(\mathcal{E})$ is necessarily many-to-one and lossy.

4.3 Axiom 3 — The Inversion

Verbal: Modernity inverted the proper order. Event → measurement became measurement → what counts as the event.

Formal Statement: $\text{Inversion}(i) := C_i(\mathcal{E}) \succ \mathcal{E}$

Interpretation: Coordinate representation has governing authority over the event.

4.4 Axiom 4 — Hierarchized Cartesian Distinction

Verbal: Cartesian distinctions between kinds of reality were turned into a hierarchy of authority. The error was the hierarchy, not the distinction.

Formal Statement:

Let $D : \mathcal{E} \to \{\mathcal{E}_1, \mathcal{E}_2, \ldots\}$ be a distinction map, and $H : D(\mathcal{E}) \to \text{priority order}$ a hierarchization operator. $H(D(\mathcal{E})) = \text{"measurable aspect has absolute priority"}$

Interpretation: Measurable aspects are elevated; lived, relational, first-person aspects are demoted. The error is not distinction itself, but authority hierarchy.

4.5 Axiom 5 — Systemic Inheritance

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Verbal: Science, institutions, and technical systems inherit and enforce this inverted hierarchy.
Formal statement: $\forall S \in \mathcal{S} : S \models C_i(\mathcal{E}) \succ \mathcal{E}$
Interpretation: Each major social subsystem encodes and reproduces the inversion.

4.6 Axiom 6 — Non-Universal Measurability of Primitive Events

Verbal: Primitive events $\mathcal{E}$ possess non-universal measurability by ontological necessity. There is no universal coordinate chart C_{univ} such that $C_{univ}(\mathcal{E}) \equiv \mathcal{E}$ for all observers. Measurability is inherently local, relational, and observer-dependent.
Formal Statement:
$\nexists C_{univ} : \mathcal{E} \to M_{univ}$ such that $C_{univ}(\mathcal{E}) \equiv \mathcal{E} \;\; \forall \mathcal{E}$
$\forall i : \dim(C_i(\mathcal{E})) < \dim(\mathcal{E})$
Consequences:

1. **All coordinate systems are partial and biased**
   Any mapping $C_i(\mathcal{E})$ is information-compressing, shaped by what the system cares about
   - Information-compressing
   - shaped by what the system cares about
   - unable, in principle, to carry the entire "transformative reality" of the event.
2. **Measurability is relative and relational**
   What can be measured depends on:
   - the observer/system
   - the instruments and models
   - the authorized language and categories
     There is no "view from nowhere" that measures *everything that matters*.
3. **Incompleteness is structural, not accidental.**
   Scientific and institutional failures are not just "implementation bugs"; they stem from the logical limits of any coordinate-based approach

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Connection to GR: Proper time and simultaneity are already observer-relative; MCT extends this to the measurability of events themselves.

Connection to RQM:

- **Relational state:** $|\psi\rangle_{S|O_i} \in \mathcal{H}_S$
- **Born rule (relative frame):** $P(a)_{S|O_i} = |\langle a|\psi\rangle_{O_i}|^2$
- **No absolute state:** $\nexists\, |\psi\rangle_{S|univ}$ with absolute meaning

4.7 Corollaries of Axiom 6

- No "view from nowhere"
- Coordinate systems are inherently lossy
- Relativity of measurability
- Horizons as relational boundaries
- Universal collapse / global resolution as suspect

4.8 Corollary 6.1 — Memory Divergence as Evidence of Non-Exhaustibility

Verbal: If any universal coordinate system C_{univ} fully captured the structure of $\mathcal{E}$, then all observers exposed to the same measured input would produce identical internal reconstructions. Empirically, they do not.

- **Formal argument:**
- **Assume for contradiction:** $C_{univ}(\mathcal{E}) \equiv \mathcal{E}\ \forall \mathcal{E}$
- **Then:** $R_i(C_{univ}(\mathcal{E})) = R_j(C_{univ}(\mathcal{E}))$
- **Observation:** $R_i(C_{univ}(\mathcal{E})) \neq R_j(C_{univ}(\mathcal{E}))$
- Let $\mathcal{E}_i^{\perp} = \mathcal{E} \setminus C_{univ}(\mathcal{E}) \neq \emptyset$
- **Reconstruction is actually:** $R_i(\mathcal{E}) = R_i(C_{univ}(\mathcal{E}),\ \mathcal{E}_i^{\perp})$
- Since $\mathcal{E}_i^{\perp} \neq \emptyset$, the assumption is contradicted.
- $\therefore \nexists\, C_{univ}$ such that $C_{univ}(\mathcal{E}) \equiv \mathcal{E}\ \forall \mathcal{E}$ ■

Interpretation: $\mathcal{E}_i^{\perp} = D_{missing}(i)$. If measured reality were full reality, memory would be uniform. It isn't. Therefore, measured reality is not full reality.

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5. Propositions and Mechanisms

Each proposition follows: Name → Verbal claim → Formal expression → Consequence/Interpretation

5.1 Proposition 1 — Governing Ontology

$G_{sys}(\mathcal{E}) = f(C_i(\mathcal{E}))$

Systems govern authorized reconstructions, not primitive events.

5.2 Proposition 2 — Ontological Filter

What kinds of reality may enter: $F_{\text{ont}} : \mathcal{E} \to \mathcal{E}_{\text{admissible}} \subset \mathcal{E}$

Reality outside $\mathcal{E}_{admissible}$ is set to zero.

5.3 Proposition 3 — Authority Filter

Who may introduce governing reality: $F_{auth} : \mathcal{E} \times O \to \mathcal{E}_{governing}$

Typically: $F_{auth}(\mathcal{E}, O_{subject}) \approx 0$, $F_{auth}(\mathcal{E}, O_{institution}) \approx \mathcal{E}_{admissible}$

Operating principle: What the subject says is subjective. What the record says is real. What the institution acts on is true.

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5.4 Proposition 4 — Interpretive Filter
Reality is reconstructed: $I_{int} : \mathcal{E} \to \mathcal{E}_{reconstructed}$
In both brains and machines, I_{int} is architecture-dependent and lossy.
5.5 Proposition 5 — Regime of Bounded Correctness
Correctness is evaluated only inside the representation:
$K_i(\mathcal{E}) \iff K_i(C_i(\mathcal{E}))$
Closed admissibility loop:

- Interpretation defines admissibility
- Admissibility defines correctness
- Correctness reinforces the interpretation

A system does not discover correctness; it enforces the conditions under which something may be called correct.

5.6 Proposition 6 — Named Reality Without Power
$A(\mathcal{E}) \wedge \neg G(\mathcal{E})$
A reality can be named, recorded, acknowledged, and still remain powerless.

5.7 Proposition 7 — Cross-Domain Recurrence
$\Pi(D) = F^D_{ont} \circ F^D_{auth} \circ I^D_{int}$
The same composite structure produces the same failure across every domain.

5.8 Proposition 8 — The Missing Dimensions
Primitive events contain a nonempty excluded remainder:
$\mathcal{E} = C_i(\mathcal{E}) \cup D^{(i)}_{missing}, \quad D^{(i)}_{missing} = \mathcal{E} \setminus M_i \neq \emptyset$

Contents of $D_{missing}$:

- Lived transformation
- Relational context
- Hidden cost
- Historical consequence
- Persistent corrected memory
- Cognitive load
- Fear, grief, dependency, and care burden

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Properties: relational, context-dependent, non-universal, cannot be collapsed back into one global grid without reinversion

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5.9 Consequences of Proposition 8

Show how missing dimensions matter in:

- QM
- Gravity
- AI
- Institutions
- Chronic illness and caregiving

6. Core Definitions, Operators, Dynamics, and Quantification

Each item follows: name → equation → variable definitions → interpretation → where used

6.1 Definition 1 — Primitive Event (Relational Event Definition)

A primitive event actualizes for observer O_i with respect to system S iff:

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1. there is physical interaction

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2. observer-relative state change is nonzero

$$\mathcal{E}_{O_i|S} \iff \mathrm{Int}(O_i, S) \ \wedge \ \Delta_{O_i}(S) \neq 0$$

Where:

- $\mathrm{Int}(O_i, S) :=$ physical coupling (energy/information exchange)
- $\Delta_{O_i}(S) := C_i(S)_{after} - C_i(S)_{before} \neq 0$

Four cases:

- **Case 1:** no interaction, no state change → no event
- **Case 2:** no interaction, state change by report → no event (belief update only)
- **Case 3:** interaction, no state change → no event (photon through glass)
- **Case 4:** interaction + state change → $\mathcal{E}_{O_i|S}$ actualizes

State observer-relativity explicitly: $\mathcal{E}_{O_i|S} \wedge \neg\, \mathcal{E}_{O_j|S}$ is permitted.

Connection to actualization threshold:

$\Phi(\mathcal{E}, O_i) = D_{KL}(p_i^{post} \| p_i^{prior})$

$\Phi \geq \Phi_{thr}(i) \Rightarrow \mathcal{E}$ registered as governing for O_i

What this resolves: Faglia's critique (2025), Pienaar's no-go theorems (2021), Lewis's dilemma (2024).

Used in: Section 7 (physics), Section 8 (relational frames), Section 9 (worked examples), Section 10 (AI)

6.2 Definition 2 — Interaction Operator

$I_{ij} : M_i \times M_j \to M_{ij}$

$I_{ij}(C_i(\mathcal{E}),\ C_j(\mathcal{E})) = C_{ij}(\mathcal{E})$

Properties:

1. **Non-commutativity:** $I_{ij} \neq I_{ji}$ in general
2. **Lossiness:** $\dim(C_{ij}) < \dim(\mathcal{E})$ but $\dim(C_{ij}) \geq \max(\dim(C_i), \dim(C_j))$
3. **Non-preexistence:** $C_{ij}(\mathcal{E})$ does not exist prior to I_{ij}
4. **Asymmetric update:** $\|\Delta_{O_i}\| \gg \|\Delta_{O_j}\|$ common
5. **Iterability:** $\lim_{n\to\infty} \|C_i^{(n)} - C_j^{(n)}\| \to \epsilon > 0$ (guaranteed by Axiom 6)

Decomposition:

- **Step 1 — Exchange:** X_{ij}
- **Step 2 — Reconstruction:** $R_i,\ R_j$
- **Step 3 — Joint description:** $C_{ij}(\mathcal{E}) = C_i'(\mathcal{E}) \cap C_j'(\mathcal{E})$

Used in: Section 7, Section 8, Section 9, Section 10

6.3 Relational Actualization Condition

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$\Phi(\mathcal{E}, O_i) = D_{KL}(p_i^{post} \| p_i^{prior})$
$\Phi \geq \Phi_{thr}(i) \Rightarrow \mathcal{E}$ actualizes for O_i
The threshold $\Phi_{thr}(i)$ is shaped by the four filters.
Used in: Section 7 (RQM, Wigner), Section 9 (worked examples)
Interpretation: the threshold is shaped by the four filters.

6.4 Persistent Correction Operator (AI)

$M_{t+1}^{(i)} = M_t^{(i)} \oplus \alpha(i)\, \hat{C}(\mathcal{E}_{ground}^{(t)})$

- $\alpha(i) = 0$: stateless LLMs (corrections vanish)
- $\alpha(i) > 0$: persistent architectures (events reshape the model)

Used in: Section 10 (AI)
Interpretation:

- $\alpha = 0$: stateless predictor
- $\alpha > 0$: persistent corrected memory architecture

6.5 Information-Theoretic Extension

$$\sum_i I(C_i(\mathcal{E});\ D_{missing}^{(i)}) \to 0$$

Dynamical reading:

$$\lim_{T\to\infty} \sum_i I(C_i(\mathcal{E});\ D_{missing}^{(i)})\Big|_{M_T^{(i)}} = 0$$

given sufficiently many grounding events and $\alpha(i) > 0$.

6.6 Governing-Authority Loss Functional

$L_i(\mathcal{E}) = d(\mathcal{E}, C_i(\mathcal{E})) \cdot w_i(\mathcal{E})$
d: distance between primitive event and coordinate representation
w_i: weight of the event (stakes, severity)
Small L_i: close representation or low stakes
Large L_i: big mismatch + high stakes → dangerously "correct" while failing the event
Interpretation: high mismatch + high stakes = dangerous bounded correctness.

6.7 Missing-Coordinate Theorem / Boxed Result

Place the theorem here as the central formal result.

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6.8 Fiber-Bundle Formulation

- B_i: base space (coordinate events)
- F_i: fiber (missing dimensions attached to each coordinate event)
- E_i: total space
- $\pi_i : E_i \to B_i$: projection

Trivial section: inversion (projection discards fiber)
Non-trivial section: correction (fiber dimensions restored to governance)

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7. Physics of Primitive Events

7.1 GR as Precedent for Non-Universal Measurability

- Proper time τ is observer-dependent; there is no universal "now"
- Simultaneity is frame-dependent
- GR already relativizes temporal measurement
- **MCT extends this:** not just time, but the measurability of events themselves is local and relational

7.2 Quantum Measurement Problem as Coordinate Inversion

- Standard QM treats the wavefunction $|\psi\rangle$ as primary reality
- Definite outcomes (what actually happens) are treated as the thing that needs explaining
- **This is ontological inversion:** the coordinate description (wavefunction) is treated as more real than the primitive event (the definite outcome)
- The measurement problem is what happens when you invert event and representation and then try to recover the event

7.3 Relational Quantum Mechanics

7.3.1 What RQM Already Says

Quantum states and facts exist only relative to an interacting system
No absolute, universal quantum state of the universe
Relational state: $|\psi\rangle_{S|O_i} \in \mathcal{H}_S$
Born rule (relative frame): $P(a)_{S|O_i} = |\langle a|\psi\rangle_{O_i}|^2$
Cross-perspective consistency is handled by requiring physical interactions for comparison

7.3.2 Faglia's Identified Gap

Faglia (2025): RQM claims events occur whenever interactions occur, but underspecifies what counts as an "event-making interaction"

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The theory says "interaction" without defining which interactions produce actualized events and which do not
This leaves the ontology incomplete: when does superposition end? When does a fact become a fact?

7.3.3 Your Direct Event Definition Relative to RQM

Insert Definition 1 here:

$$\mathcal{E}_{O_i|S} \iff \mathrm{Int}(O_i, S) \ \wedge \ \Delta_{O_i}(S) \neq 0$$

This is the MCT solution to the Faglia gap: an event actualizes for O_i if and only if there is physical coupling AND observer-relative state change. Not all interactions produce events. Only those that change the observer's description.

7.3.4 Relational Actualization as Solution

Insert Φ framework:

$$\Phi(\mathcal{E}, O_i) = D_{KL}(p_i^{post} \| p_i^{prior})$$

$$\Phi \geq \Phi_{thr}(i) \Rightarrow \mathcal{E} \text{ actualizes for } O_i$$

This provides the missing threshold: not every interaction-plus-state-change is equally event-making. The actualization functional quantifies how strongly the event registers.

7.3.5 Cross-Perspective Alignment via Interaction Operator

Use I_{ij} here:

- Two observers with different relational facts can only compare via physical interaction
- The interaction operator produces a joint description $C_{ij}(\mathcal{E})$ that did not exist before
- Shared reality is emergent, not pre-given
- Residual disagreement $\epsilon > 0$ is guaranteed by Axiom 6

7.4 QBism as Neighboring Partial Move

- QBism also rejects absolute quantum states and centers the agent
- QBism assigns probabilities to an agent's experiences, not to observer-independent reality
- **Contrast:** QBism risks solipsism (each agent's experience is private and non-shareable)

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- MCT avoids this via the interaction operator: shared reality is constructed through interaction, not assumed or denied
- QBism empowers the subject but provides no mechanism for cross-perspective alignment

7.5 RQM in Quantum Gravity and LQG

- **Background independence:** spacetime itself is not a fixed stage but emerges from relational dynamics
- In LQG, geometry is quantized; spacetime points are not fundamental
- **This supports the MCT claim:** spacetime coordinates are representations, not primitive reality
- Spin networks and spin foams as relational structures, not coordinate grids

7.6 Black-Hole Cosmology and Einstein–Cartan

7.6.1 Torsion Bounce

- **Einstein–Cartan theory:** spacetime has torsion (from spin density), not just curvature
- **Popławski:** torsion prevents singularity formation; collapsing matter bounces
- Instead of a singularity, the interior of a black hole may undergo a torsion-mediated bounce, producing a new expanding region
- The singularity is not a physical event but an artifact of coordinate incompleteness

7.6.2 Interior Cosmology / Parent Black Hole Model

- Our universe may exist inside a black hole formed in a parent universe
- The Big Bang corresponds to the torsion bounce inside the parent black hole
- Cosmological horizon as relational boundary: what is "inside" depends on which observer

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7.6.3 Supermassive Black Holes as Stabilizers

- SMBHs at galactic centers as anchors for galactic structure
- **AGN feedback:** energy output regulates star formation, gas dynamics
- **Self-regulating systems:** the black hole shapes its environment, which shapes accretion, which shapes the black hole

MCT reading: the SMBH is not just a gravitational object but a relational node whose "event" includes the entire galaxy's co-evolution

7.6.4 JWST / JADES Axis / Rotation Asymmetry

- JWST and JADES observations of unexpected structure at high redshift
- Galaxies more massive and more structured than standard models predicted at early epochs
- Possible asymmetry / preferred axis in galaxy rotation statistics

MCT reading: these anomalies may indicate that the standard cosmological coordinate framework (ΛCDM parameters) is missing dimensions of the primitive cosmological event

7.7 Bounded Correctness in Physics

- Physics is powerful but structurally incomplete within its coordinates
- The measurement problem, the black-hole information paradox, and the QM–GR gap are all instances of bounded correctness: the formalism is internally consistent but fails the primitive event
- This is not a criticism of physics; it is an application of the theory to physics

8. Relational Frames, Interaction, and Emergent Shared Reality

8.1 Formal Relational Model

- Observers O_i, O_j with coordinate maps C_i, C_j
- Each observer has a partial, lossy representation of $\mathcal{E}$
- No pre-given shared description exists

8.2 Shared Reality as Process

- Objectivity is not a property of a description; it is a process of alignment through interaction
- I_{ij} fires when both observers interact and both undergo state change
- The joint description $C_{ij}(\mathcal{E})$ is the result, not the precondition

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8.3 Pattern vs. Actual Event

Three distinct ontological states:

- **Pattern / possibility:** an event type that could occur but has not actualized for any observer (e.g., "we might have tacos")
- **Actualized but unknown:** the event has actualized for O_i but O_j does not yet know (e.g., the friend has measured spin; Wigner has not)
- **Actualized and shared:** interaction has occurred and $C_{ij}(\mathcal{E})$ exists

8.4 Alignment Interactions as First-Class Events

- The act of aligning frames is itself a primitive event
- When I_{ij} fires, both observers are changed
- The alignment event has its own missing dimensions (what was lost in translation, what power asymmetry shaped the exchange)

8.5 Dinner / Taco Tuesday

- **Setup:** Partner A decides to make tacos. Partner B doesn't know yet.
- t_0: Decision exists as pattern/possibility
- t_1: A acts (shops, cooks) — event actualizes for A but not for B
- t_2: B walks in, smells tacos — I_{AB} fires, joint description $C_{AB}(\mathcal{E})$ created
- **Missing dimensions:** A's effort, planning, compromise; B's surprise, gratitude or indifference
- **Bounded correctness:** A calendar entry ("Dinner: Tacos, 6 PM") is correct and misses everything

8.6 Restaurant Choice

- Distinguishes "not-yet-actualized" (we haven't decided where to eat) from "actualized-but-unknown" (I chose the restaurant but you don't know yet)
- Decision as event: the moment of choosing is an actualization for the chooser
- Communication as interaction: telling the other person fires I_{ij}

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- Residual disagreement: even after alignment, each person's experience of "dinner" differs

8.7 Wigner-Style Divergence

- **Inside the lab:** friend has a definite outcome (event actualized)
- **Outside the lab:** Wigner's description is superposition (event not actualized for Wigner)
- This is not a paradox under MCT — it is the expected consequence of Definition 1
- The friend has $\mathcal{E}_{F|S}$; Wigner does not have $\mathcal{E}_{W|S}$ until interaction

8.8 Horizon Case

- **Infalling observer:** crosses the horizon, event actualizes (tidal forces, time dilation experienced)
- **Distant observer:** sees the infalling observer freeze at the horizon (event never actualizes in their frame)
- Same physical situation, two different event structures
- **Missing dimensions:** the infalling observer's experience is not captured by the distant observer's coordinate description
- The horizon is a relational boundary, not an absolute one

9. Worked Examples

For each example, use the same 7-step template:

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1. Identify the primitive event $\mathcal{E}$
2. Identify each observer/system O_i
3. Write the coordinate maps $C_i(\mathcal{E})$
4. Identify what is missing: $D^{(i)}_{missing}$
5. Identify whether actualization occurred: $\Phi \geq \Phi_{thr}$?
6. Identify whether interaction aligned frames: I_{ij} fired?
7. Identify where bounded correctness appears: $K_i(C_i(\mathcal{E}))$ true but $K_i(\mathcal{E})$ false?

9.1 Wigner's Friend — Full Three-Phase Walkthrough
Setup:

- **System** S**:** quantum system (e.g., spin-$\frac{1}{2}$ particle)
- **Friend** F**:** inside the lab, measures S
- **Wigner** W**:** outside the lab

Phase 1: t_0 (before measurement)
Neither F nor W has interacted with S
$\neg\,\mathcal{E}_{F|S}$ and $\neg\,\mathcal{E}_{W|S}$
Both assign prior distributions

Phase 2: t_1 (friend measures)
$\mathrm{Int}(F, S)$: physical coupling occurs
$\Delta_F(S) \neq 0$: friend's description changes (e.g., "spin up")
$\mathcal{E}_{F|S}$ actualizes
$\Phi(\mathcal{E}, F) \geq \Phi_{thr}(F)$
For Wigner: $\neg\,\mathrm{Int}(W, S)$ and $\Delta_W(S) = 0$
No paradox: the event is real for the friend, not real for Wigner. This is not contradiction; it is observer-relative actualization.

Phase 3: t_2 (Wigner opens the door)
$\mathrm{Int}(W, F)$: physical interaction
$\Delta_W(F) \neq 0$: Wigner's description changes
I_{FW} fires
Joint description $C_{FW}(\mathcal{E})$ is created
But: $\dim(C_{FW}(\mathcal{E})) < \dim(\mathcal{E})$

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Missing dimensions remain: the friend's experience of being inside the lab during the measurement, the temporal order of actualization, the asymmetry of who knew first

Summary table:

Time	Friend	Wigner	Joint
t_0	no event	no event	—
t_1	$\mathscr{E}$_F|S actualizes	no event	—
t_2	updated	$\mathscr{E}$_W|F actualizes	C_FW($\mathscr{E}$) created via I_FW

Bounded correctness: Wigner's quantum-mechanical description at t_1 (superposition of friend + system) is correct inside Wigner's coordinate space and wrong relative to the friend's actualized event.

9.2 Dinner / Taco Tuesday — Applied Everyday Case

1. **Primitive event:** A decides to cook tacos for the family
2. **Observers:** O_A (cook), O_B (partner), O_C (child)
3. **Coordinate maps:**
   - C_A: ingredients bought, recipe followed, time spent, physical effort
   - C_B: "tacos for dinner", discovered upon arrival
   - C_C: "yay tacos" or "I don't want tacos"
4. **Missing dimensions:**
   - A's fatigue, planning burden, compromise (wanted pasta but B hates pasta)
   - B's ignorance of A's effort
   - C's emotional state unrelated to food
5. **Actualization:**
   - $\Phi(\mathcal{E}, A) \gg \Phi_{thr}(A)$ (A lived the full event)
   - $\Phi(\mathcal{E}, B) \geq \Phi_{thr}(B)$ only after walking in
6. **Interaction:** I_{AB} fires at dinner; C_{AB} = shared experience of the meal
7. **Bounded correctness:** A family calendar entry ("Tues: Tacos") is correct and captures almost nothing

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9.3 Black-Hole / Horizon Case — Applied Cosmological Case

1. **Primitive event:** matter crosses a black-hole horizon
2. **Observers:** O_{in} (infalling), O_{dist} (distant)
3. **Coordinate maps:**
   - C_{in}: smooth passage, tidal forces, proper time continues
   - C_{dist}: object freezes at horizon, redshifts to invisibility
4. **Missing dimensions:**
   - For O_{dist}: the infalling observer's continued experience
   - For O_{in}: the distant observer's perception of freezing
   - For both: the interior geometry (if Einstein–Cartan torsion bounce occurs)
5. **Actualization:** $\mathcal{E}_{O_{in}|horizon}$ actualizes; $\mathcal{E}_{O_{dist}|horizon}$ does not (the event never registers as a definite crossing in the distant frame)
6. **Interaction:** $I_{in,dist}$ cannot fire across the horizon (one-way causal boundary)
7. **Bounded correctness:** Both GR descriptions are internally consistent and describe different events. Neither is wrong within its frame. Both are incomplete.

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10. Machine Cognition and AI

10.1 Reasoning as Memory Applied Forward

- **Damasio:** somatic markers ground reasoning in bodily memory
- **Friston:** predictive processing — the brain minimizes prediction error by updating its generative model
- Reasoning is not abstract logic; it is memory (accumulated experience) applied to new situations
- Without persistent, corrected memory, there is no reasoning — only pattern-matching on a frozen training distribution

10.2 Human–Machine Convergence

- Both brains and AI systems are predictive systems
- Both compress events into coordinate representations
- Both can exhibit bounded correctness (internally consistent, externally wrong)
- **The difference:** brains update continuously from grounded experience; current AI does not

10.3 Architectural Amnesia in Current AI

- Stateless architecture: each session starts from zero
- User corrections vanish at session end
- The system cannot learn that it was wrong
- Every interaction is a first interaction

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- This is not a bug to be patched; it is the architectural expression of $\alpha(i) = 0$

10.4 The Inversion in AI

- The model's statistical baseline (trained on population-level data) governs output
- The user's specific, corrected, grounded reality does not persist
- **This is ontological inversion:** the coordinate representation (training distribution) governs the event (the user's actual situation)
- The AI is "correct" relative to its training data and wrong relative to the user's primitive event

10.5 Proposition — The Missing Coordinate in Machine Cognition

- **Claim:** The missing coordinate in current AI is persistent, grounded, corrected memory.
- Not just "memory" (context windows, retrieval-augmented generation)
- **Persistent:** survives across sessions
- **Grounded:** tied to actual user events, not statistical priors
- **Corrected:** when the user says "that's wrong," the correction accumulates

10.6 Persistent Correction Operator — Full Form

$$M_{t+1}^{(i)} = M_t^{(i)} \oplus \alpha(i)\, \hat{C}(\mathcal{E}_{ground}^{(t)})$$

Where:

- $M_t^{(i)}$: memory state of system i at time t
- $\mathcal{E}_{ground}^{(t)}$: a grounding/correction event at time t
- $\hat{C}(\mathcal{E}_{ground}^{(t)})$: encoded correction
- $\alpha(i) \in [0, 1]$: persistence/learning rate
- $\oplus$: structure-preserving update

Interpretation:

- $\alpha = 0$: current stateless LLMs. Corrections don't survive. No reasoning.
- $\alpha > 0$: architectures where events gradually reshape the model. Reasoning becomes possible.
- $\alpha \to 1$: full persistence — every correction permanently alters future behavior.

10.7 Stateless vs. Reasoning AI Comparison Table

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Feature	Stateless AI ($\alpha = 0$)	Reasoning AI ($\alpha > 0$)
Memory	Session-only; resets	Persistent across sessions
Corrections	Vanish at session end	Accumulate and govern
User model	Population-level statistical prior	Grounded, corrected, user-specific
Error pattern	Same mistake every session	Mistakes decrease over time
Authority	Model governs user reality	User governs model reality
Ontological status	Coordinate system (C_i) is fixed	Coordinate system updates toward $\mathcal{E}$
Bounded correctness	Always: correct inside training distribution, wrong for user	Decreasing: $L_i(\mathcal{E}) \to 0$ as $t \to \infty$
MCT diagnosis	Missing coordinate = no persistent corrected memory	Missing coordinate partially restored

10.8 Architectural Implications

- Persistent corrected memory is necessary but not sufficient

Must also address:

- Who owns the memory (authority filter → user sovereignty)
- What can be silently reinterpreted (ontological filter → canonical locked objects)
- How corrections enter the system (interpretive filter → deterministic constraint layer before probabilistic generation)
- Whether the system monitors its own drift (bounded correctness → drift detection)
- This is the bridge to the Bodhi Systems architecture (if included as design application)

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10.9 AI as Bounded Correctness in Live Form

- Current AI is the clearest, most visible, most reproducible demonstration of bounded correctness
- Every user who has corrected an AI and watched the correction vanish has experienced the theory's core claim
- The AI is procedurally perfect (generates fluent, coherent, statistically probable text) and structurally false (to the user's actual, corrected, grounded reality)
- This makes AI not just an application domain for the theory but a proof-of-concept environment where the theory's predictions are directly testable

11. Filters and the Architecture of Bounded Correctness

11.1 The Four Filters

Present the complete filter pipeline as a unified mechanism:

Filter 1 — Ontological Filter

$F_{ont} : \mathcal{E} \to \mathcal{E}_{admissible}$

- Determines what kinds of things are allowed to count as real in the system

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- If something cannot be formally defined (relational context, chronic pain, trauma, care burden), it is treated as non-existent for decision purposes
- Reality outside $\mathcal{E}_{admissible}$ is set to zero

Filter 2 — Authority Filter

$F_{auth} : \mathcal{E} \times O \to \mathcal{E}_{governing}$

- Determines who is allowed to introduce reality in governing form
- Creates the system-subject: the person closest to the event whose description is treated as "subjective"
- Operating principle: "What the subject says is subjective. What the record says is real. What the institution acts on is true."

Filter 3 — Interpretive Filter

$I_{int} : \mathcal{E} \to \mathcal{E}_{reconstructed}$

- Transforms incoming reality into the language and categories the system can process
- A lived medical event enters as a diagnostic code; a legal situation enters as a charge; a user correction enters as a token sequence
- Architecture-dependent and lossy in both brains and machines

Filter 4 — Correctness Filter

$K_i(\mathcal{E}) \iff K_i(C_i(\mathcal{E}))$

- The system evaluates correctness only within its own coordinates
- A system is "correct" when its outputs are consistent with its chosen coordinate space
- This says nothing about whether it is faithful to the primitive event

Composite filter:

$\Pi(D) = F^D_{ont} \circ F^D_{auth} \circ I^D_{int}$

11.2 Closed Admissibility Loop

How admissibility self-reinforces:

- Interpretation defines admissibility
- Admissibility defines correctness
- Correctness reinforces the interpretation
- The loop is closed: the system cannot discover its own failure from inside its own coordinates
- A system does not discover correctness; it enforces the conditions under which something may be called correct

11.3 Governing-Authority Loss Functional in Mechanism Form

$L_i(\mathcal{E}) = d(\mathcal{E}, C_i(\mathcal{E})) \cdot w_i(\mathcal{E})$

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Apply here as the quantitative measure of how badly a system fails an event after the filters have operated:

- d measures representational mismatch (how far the coordinate event is from the primitive event)
- w measures stakes (how much it matters)
- A system with high L_i is dangerously "correct" — internally consistent, high-stakes failure

11.4 Named Reality Without Power

$A(\mathcal{E}) \wedge \neg G(\mathcal{E})$

Applied examples:

- A patient's pain is documented in the chart but does not change the treatment plan
- A user's correction is acknowledged by the AI ("You're right, I apologize") but does not persist to the next session
- A legal precedent is cited but does not alter the ruling
- A policy report names a problem but no budget line is created

11.5 Why Systems Become Right About What They Allow Themselves to See

Closing mechanism claim:

- Bounded correctness is not a bug; it is the stable attractor of any system that governs through coordinate representations
- The system is not lying; it genuinely cannot see what its coordinates exclude
- Correction requires breaking the admissibility loop from outside — restoring the primitive event to governing authority
- This is what the theory calls "inverting the inversion"

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12. Institutional and Civilizational Applications

12.1 Medicine

- The chart outranks the patient
- Lab values are admissible; lived illness is not
- A diagnostic code travels through the system; the event that generated it does not
- Procedures extract blood, CSF, scans, time, pain, and hope — and often return "normal" results or dismissal
- The patient is the system-subject: closest to the event, least authorized to define it
- Bounded correctness: the system is procedurally active and structurally false to the event

12.2 Law and Bureaucracy

- The case file outranks the situation
- Eligibility criteria define admissible reality; need does not
- A person's legal situation enters the system as a charge, a filing, a form
- The form travels; the person does not
- Procedural compliance is correctness; justice is outside the frame

12.3 Governance and Metrics

- GDP, KPIs, risk scores define what counts as real for governance
- The metric governs; the event the metric was supposed to measure does not
- A city's "improving" statistics can coexist with worsening lived conditions
- The metric is correct inside its own coordinates

12.4 Academia and Knowledge Production

- Peer review enforces admissibility (what counts as valid method, valid question, valid contributor)
- Method filters determine what kinds of evidence are real
- Outsider exclusion: work produced outside institutional coordinates is rendered non-governing regardless of content
- Publication venue outranks argument quality as a credibility signal

12.5 System-Subject and Authority Asymmetry

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"What the subject says is subjective. What the record says is real. What the institution acts on is true."

- The system-subject is the person closest to the primitive event
- Their testimony is downgraded by the authority filter
- The institutional record carries governing authority
- This asymmetry is not incidental; it is structural and reproducible across every domain

12.6 Institutional Bounded Correctness — Synthesis

- Every institution listed above exhibits the same composite filter $\Pi(D)$
- Each is internally consistent
- Each fails the primitive event in the same structural way
- The theory does not claim institutions are malicious; it claims they are incomplete and that incompleteness is systematically invisible from inside

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13. Phenomenological and Empirical Case Studies

This section presents "Living Without Autopilot" as a core case of the theory's claims in lived form. It is not memoir appended to theory; it is phenomenological evidence.

13.1 Empirical Case: Living Without Autopilot

The essay documents:

- Collapse of predictive compression
- Breakdown of implicit sequencing
- Manual event decomposition
- Hidden costs outside formal measurement
- Need for external corrective architecture

Theory-language bridge sentence:

"Living Without Autopilot *documents the collapse of predictive compression at the level of lived experience: actions that ordinarily function as unified events become manually decomposed into sequentially governed sub-events, revealing the hidden cost structure and missing dimensions that coordinate systems fail to capture."*

13.2 Loss of Autopilot as Event Decomposition

Key passages to map:

- *"There is no autopilot to carry me"*
- *"Life is no longer a fluid experience but a series of fragmented tasks"*
- *"every 'simple' thing takes ten extra steps my brain won't let me skip"*

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Map to:

- Fragmented event structure: what was a single unified event ("take your medication") decomposes into a cascade of sub-events, each requiring conscious resolution
- Collapse of abstraction: healthy cognition compresses multi-step sequences into single events; chronic illness forces explicit resolution of every intermediate state
- Forced manual governance: each sub-event must be individually actualized, tracked, and completed

13.3 The Brutal Economy of Pain as Missing Dimensions

Map the essay's "ledger" to $D_{missing}^{(i)}$:

- Pain (constant, unending tax)
- Energy (a shower costs the ability to cook dinner)
- Cognitive load (decision fatigue beyond what the term captures)
- Fear (of the next crash, of forgetting a step)
- Guilt (at needing help)
- Uncertainty (every test should buy answers; too often buys shrugs)
- Care burden (radiating to family, children, relationships)

Theory claim: current systems do not measure the true cost structure of the event.

The "Healthy Economy" vs. "Chronic Illness Economy" table in the essay is a direct instantiation of $D_{missing}$:

- **Doctor's visit:** co-pay vs. vials of CSF, days of agony, scars
- **Birthday party:** buying a gift vs. the ability to function for two days
- **Taking medication:** a sip of water vs. cognitive load, physical effort, fear of side effects

13.4 A Passport Stamped in Scars as Ontological Fragmentation

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Map the medical-specialty fragmentation to ontological inversion:

- *"I am not seen as a whole person, but as a collection of disparate cases"*
- Neurology stamped with electrodes; Rheumatology with *"unclear"*; Pulmonology with oxygen measures; Hematology with clotting risks; Gastroenterology with scopes; Psychiatry with questions that reduce body failure to mind failure
- Multiple coordinate systems $C_1, C_2, \ldots, C_n$ each indexing partial aspects of one primitive event $\mathcal{E}$
- None exhausts or governs the whole
- The patient is a *"permanent, stateless refugee in the land of the sick"*
- The passport is proof of system-subject disassembly

13.5 The Feedback Loop as Closed-Loop Instability

Map the rage/guilt/invalidation cycle:

- Pain → effort → system dismissal → rage → guilt → shame → renewed effort → repeated failure
- *"This is the feedback loop from which there is no easy escape"*
- In theory terms: a closed-loop instability produced when the system fails to resolve the event but continues to govern it through partial representations
- The system's dismissal is the ingredient that turns physical pain into a psychological prison
- This is not emotional weakness; it is a structural consequence of bounded correctness applied to a person

13.6 The Hardest Symptom as Relational Catastrophe

Map the dependency/child/family sections:

- *"The hardest symptom of all is needing help"*
- *"I saw the fear bloom in her eyes"*
- **The illness's true malignancy:** ability to radiate suffering outward, to become a relational catastrophe
- **Missing relational dimensions:** the primitive event exceeds the biomedical coordinate frame
- The biomedical model defines symptoms as observable phenomena within the patient; the theory says the event includes the fear in a child's eyes
- **This is Proposition 8 in its most visceral form:** $D_{missing}$ includes the relational, familial, and intergenerational consequences

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13.7 Survival, Not Perfectionism: Manual Architecture

Map the final section on building detailed systems:

- *"If I don't have every detail written down — the exact pill, the precise dose, the specific time — someone could miss a step. And in my world, a missed step could cost me my life."*
- **External persistence architecture:** lists, medication structures, procedural order built because internal autopilot cannot be trusted
- **Human analogue of persistent corrected memory:** $M_{t+1} = M_t \oplus \alpha \cdot \hat{C}(\mathcal{E}^{(t)}_{ground})$
- The loss of autopilot forced the manual creation of a persistent corrective system
- **This is the AI argument made flesh:** reasoning requires persistent, grounded, corrected memory — whether the system is silicon or biological
- *"That's not perfectionism. That's not control. That's survival."*

14. Scenario Dataset (Broader Illustrations)

14.1 Coordination and Expectation

Taco Tuesday, restaurant choice.

14.2 Measurement and Observation

Wigner's friend, internal vs. external frames.

14.3 Logging vs. Lived Reality

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Baby feeding at 2:17 AM. Conflicting timestamps. The log says *"fed at 2:17 AM"*; the event includes exhaustion, fear, the weight of the baby's body, the dark hallway, the question *"am I doing this right?"*

14.4 Catastrophe and Propagation
City-scale event locally real first. News coordinates register it globally; the primitive event is local, bodily, specific.

14.5 Identity and Category Pressure
Trans identity resisting binary closure. The system's coordinate categories (M/F) do not exhaust the event of identity. The person is the system-subject.

14.6 Continuity and Memory Loss
External system preserving continuity when internal memory fails. Alzheimer's, TBI, chronic illness brain fog — the event continues; the coordinate system (memory) collapses.

14.7 Stateless vs. Stateful AI
Forgetting vs. correction-preserving systems. The same correction given in session 1 must be given again in session 2, 3, 4, ..., n.

14.8 Scenario-Mapping Workflow
For each scenario, the 7-step template:

1. Primitive event $\mathcal{E}$
2. Observers/systems O_i
3. Coordinate maps $C_i(\mathcal{E})$
4. Missing dimensions $D^{(i)}_{missing}$
5. **Actualization:** $\Phi \geq \Phi_{thr}$?
6. **Interaction alignment:** I_{ij} fired?
7. **Bounded correctness:** $K_i(C_i(\mathcal{E}))$ true, $K_i(\mathcal{E})$ false?
8. **Authority asymmetry:** who is the system-subject?

15. Quantification and Geometric Upgrade
15.1 Governing-Authority Loss Functional — Full Treatment
$L_i(\mathcal{E}) = d(\mathcal{E}, C_i(\mathcal{E})) \cdot w_i(\mathcal{E})$

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Full interpretation:

- Can be used to rank systems by failure severity
- Can be used to compare domains (which system fails the same event worse?)
- **Possible normalization:** $\bar{L}_i = L_i / \max(L)$
- **Connection to information-theoretic formulation:** L_i increases as mutual information between $C_i(\mathcal{E})$ and $D^{(i)}_{missing}$ decreases

15.2 Fiber-Bundle Structure of Missing Dimensions — Full Mathematical Presentation

- **base space:** $B_i = C_i(\mathcal{E})$
- **fiber:** $F_i(C_i(\mathcal{E})) = D^{(i)}_{\text{missing}}(\mathcal{E})$
- **total space:** $E_i = \bigcup_{\mathcal{E}} (C_i(\mathcal{E}) \times F_i(C_i(\mathcal{E})))$
- **projection:** $\pi_i : E_i \to B_i$

Base space B_i:

- The space of coordinate events $C_i(\mathcal{E})$
- What the system can see and represent

Fiber F_i:

- The missing dimensions $D^{(i)}_{missing}$ attached to each coordinate event
- What the system cannot see but what is real

Total space E_i:

- $E_i = B_i \times_{\pi_i} F_i$
- The full event structure combining coordinate and missing dimensions

Projection $\pi_i : E_i \to B_i$:

- Maps from total space to coordinate base
- This is ontological inversion formalized geometrically: the system operates on B_i (the projection), discarding the fiber

Sections:

A section $s : B_i \to E_i$ assigns to each coordinate event its missing dimensions

Trivial section: the system ignores the fiber; $\pi_i \circ s = \mathrm{id}$ collapses to coordinate-only governance. This is the inversion.

Non-trivial section: the system actively models the fiber; missing dimensions are restored to governance. This is the correction.

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15.3 Geometry-to-Physics Bridge

- In GR, the fiber bundle is the tangent bundle over spacetime
- In gauge theory, the fiber carries internal symmetry degrees of freedom
- In MCT, the fiber carries event dimensions excluded by any particular coordinate chart
- The analogy is structural, not metaphorical: just as gauge fields live in fibers over spacetime, missing dimensions live in fibers over coordinate events

Connection to LQG: if spacetime itself is emergent from relational structure, then the base space B_i is itself a derivative object, and the fiber is where the relational structure that generates it lives

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16. Unified Interpretation

16.1 Synthesis of Axioms, Propositions, Definitions, and Applications

Restate the whole system cleanly:

1. Events are ontologically prior to representations (Axiom 1)
2. No representation exhausts an event (Axiom 2)
3. Modern systems invert this order (Axiom 3)
4. The inversion is inherited from the Cartesian hierarchy (Axiom 4) and propagated through institutions (Axiom 5)
5. Non-universal measurability is structural, not accidental (Axiom 6)
6. Missing dimensions are real and must be explicitly modeled (Proposition 8)
7. The four filters produce bounded correctness: systems correct inside their coordinates, false to the event
8. The interaction operator and actualization condition provide the relational mechanism
9. Persistent corrected memory is the missing coordinate in AI
10. The same structure appears in physics, AI, institutions, and lived experience

16.2 One Failure Mode Across Domains

- **Physics:** wavefunction governs; definite outcome needs explaining
- **AI:** training distribution governs; user correction vanishes
- **Medicine:** chart governs; patient experience is *"subjective"*
- **Law:** file governs; situation is *"irrelevant"*
- **Chronic illness:** system fragments the person; the unified event is nowhere in the coordinates
- **Daily life:** the calendar entry is correct and empty

All are instances of: $C_i(\mathcal{E}) \succ \mathcal{E}$

17. Final Formal Statement of the Theory

The tight four-point canonical statement:

1. **Event priority:** Primitive events $\mathcal{E}$ are ontologically prior to any coordinate representation $C_i(\mathcal{E})$.

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2. **Non-universal measurability:** No universal coordinate chart C_{univ} can exhaustively capture all primitive events for all observers. Measurability is inherently local, relational, and observer-dependent.
3. **Missing dimensions:** Every coordinate representation excludes real aspects of the event: $\mathcal{E} = C_i(\mathcal{E}) \cup D^{(i)}_{missing}$, $D^{(i)}_{missing} \neq \emptyset$.
4. **Bounded correctness under inversion:** When systems treat $C_i(\mathcal{E})$ as ontologically prior to $\mathcal{E}$, they produce a regime of bounded correctness — internally consistent, structurally false to the event.

Then the short quotable paragraph version:
"*The fundamental error of modern systems is not that they fail to measure reality, but that they mistake the measured representation of an event for the event itself. This produces systems that are correct inside their own coordinates and wrong in every way that matters. The correction is not to abandon measurement, but to restore governing authority to the events that measurement was created to serve.*"

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18. Self-Diagnosis

18.1 The Paper as Coordinate Representation

This document is itself a coordinate representation $C_{paper}(\mathcal{E}_{theory})$ of the primitive event of the theory's creation.

18.2 The Theory's Creation as Primitive Event

The event of producing this theory exceeds its representation:

- Written during chronic illness, brain fog, caregiving
- Produced using AI as a continuity keeper when sustained thought was impossible
- The theory predicted this would work; it did
- The process is itself a proof of concept for the framework it describes

18.3 Predicted Reception

Apply the four filters to the paper itself:

- **Ontological filter:** Does the academy admit work produced outside institutional coordinates? By an independent researcher? Using AI as cognitive infrastructure?
- **Authority filter:** Who is authorized to produce theory? A patient? A mother? Someone without a PhD?
- **Interpretive filter:** How will the work be recoded? Self-help? Speculative? Not rigorous?

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- **Correctness filter:** If the work is evaluated by criteria it explicitly diagnoses as incomplete, the evaluation will be bounded-correct

18.4 Refusal to Let the Paper Become the Event

- Treating this document as the total event repeats the inversion
- The document names its own incompleteness
- The primitive event includes everything this paper cannot carry: the pain, the children, the fight, the AI sessions at 3 AM, the scars

18.5 The Paper Names Its Own Incompleteness on Purpose

Strong closing line. The theory applies to itself and survives. Use verbatim.

19. Implications and Future Work

19.1 Physics

- **Relational quantum gravity:** event-first formalism as complement to LQG
- Horizon-based event frameworks
- Fuller event formalism for RQM (building on Definition 1 and the Faglia solution)
- **Empirical tests:** do cosmological anomalies (JWST/JADES) correspond to missing-dimension predictions?

19.2 AI

- Persistent corrected-memory architectures
- Event-sensitive systems that update from grounded user interaction
- Drift detection as architectural feature, not afterthought
- Bodhi Systems as design application (reference if included)

19.3 Institutions

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- **Weaken filters:** make more kinds of reality admissible
- **Restore event authority:** the person closest to the event should have governing input
- **Improve auditability:** make the filters visible and contestable
- Design systems that can detect their own bounded correctness

19.4 Medicine and Care

- Use lived-event structures in governance, not just recorded proxies
- Patient-reported outcomes as governing data, not supplementary
- Chronic illness management that accounts for the full cost structure (*the brutal economy of pain*)
- Holistic event governance instead of specialty fragmentation

19.5 Philosophy

- Shift from measurement-first to event-first ontology
- Formal engagement with phenomenology (Heidegger, Merleau-Ponty) via the event-coordinate distinction
- Formal engagement with epistemic injustice literature (Carel, Kidd, Fricker) via the authority filter

20. References

Include:

- Rovelli (RQM foundational papers)
- Adlam & Rovelli (cross-perspective consistency)
- Faglia (2025) (gap in RQM event specification)
- Pienaar (2021) (no-go theorems for RQM)
- Lewis (2024) (relational dilemma)
- Popławski (Einstein–Cartan torsion, cosmology inside black holes)
- Vidotto (black-hole cosmology, LQG)
- Dorato (philosophy of time, events)
- Damasio (somatic markers, embodied reasoning)

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- Friston (predictive processing, free energy principle)
- Persistent-memory AI papers (as identified in search)
- Standard GR / LQG sources where equations are used
- Carel & Kidd (epistemic injustice in healthcare)
- Fricker (epistemic injustice foundational)
- Author's own documents/essays where appropriate

21. Appendices

Appendix A — Full axiom set (Axioms 1–6 with all formal statements)
Appendix B — Full propositions (Propositions 1–8 with all formal statements)
Appendix C — Expanded RQM formalism and critiques (Faglia, Pienaar, Lewis; full technical engagement)
Appendix D — Einstein–Cartan/Popławski technical notes (torsion tensor, bounce conditions, interior cosmology)
Appendix E — Machine cognition supplement (full AI memory update dynamics, convergence conditions, stateless-vs-persistent architecture comparison)
Appendix F — Equations as Indexing Tools (meta-discussion of how the theory's own equations are coordinate representations of its claims)
Appendix G — Plain-language summary (the entire theory in 2–3 pages for non-specialist readers)
Appendix H — Living Without Autopilot (full text or extended excerpts, if not fully embedded in Section 13)

0. Abstract

This paper introduces The Missing Coordinate Theory (MCT), a formal framework asserting that primitive events, the full, lived, transformed occurrences that constitute reality, are ontologically prior to any coordinate representation used to measure, record, or govern them. Modern systems

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across physics, artificial intelligence, and institutional governance commit what this theory calls **ontological inversion**: they treat the coordinate representation of an event as if it were the event itself, systematically discarding aspects of reality that do not fit their admissible dimensional frameworks. We formalize this claim through six axioms, culminating in Axiom 6 (Non-Universal Measurability), which establishes that no single coordinate chart can exhaustively capture all primitive events for all observers, a principle already realized in general relativity's observer-dependent proper time and in Relational Quantum Mechanics' rejection of absolute quantum states. Proposition 8 (The Missing Dimensions) provides the constructive complement: every coordinate representation $C_i(\mathcal{E})$ of a primitive event $\mathcal{E}$ excludes a non-empty space of missing dimensions $D^{(i)}_{missing}$, including lived transformation, relational context, stakes, consequences, and persistent corrected memory. This framework is operationalized through four filters (ontological, authority, interpretive, and correctness) that produce a Regime of Bounded Correctness: systems that are internally consistent yet structurally false to the events they govern. We present formal operators, including a relational event definition that addresses an identified gap in RQM (Faglia, 2025), an interaction operator for cross-perspective alignment, and a persistent correction operator for machine cognition, and apply the theory across quantum measurement, black-hole cosmology, AI architecture, institutional failure, and the phenomenology of chronic illness, where the essay ***Living Without Autopilot*** demonstrates the collapse of predictive compression as a lived instantiation of missing coordinates.

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1. Introduction

1.1 Motivation and Central Question

This paper begins from a single observation: the same structural failure appears across domains that have no apparent connection to one another.

In quantum mechanics, the formalism treats the wavefunction, a coordinate representation of possible outcomes, as primary reality. When a definite outcome actually occurs (the primitive event), the formalism has no smooth mechanism to produce it. The definite outcome must be recovered through an ad-hoc *"collapse"* postulate, because the governing representation has been elevated above the event it was created to describe.

In black-hole physics, an observer falling through a horizon undergoes a continuous, physically real experience, tidal forces, proper time advancing, the interior geometry unfolding. A distant observer, using their own coordinate system, describes the same situation as the infalling observer freezing at the horizon and redshifting to invisibility. Both descriptions are internally consistent. Neither exhausts the primitive event. The horizon is not an absolute boundary; it is a relational one, and the two coordinate descriptions are irreconcilable without acknowledging that event reality is observer-relative.

In artificial intelligence, the statistical model trained on population-level data generates outputs by predicting what is likely given learned patterns. When a user corrects the system, *"that's not what I meant," "you're wrong about this," "my situation is different"*, the correction is acknowledged in the moment. Then the session ends. The correction vanishes. The next session begins from zero, governed again by the unrevised statistical baseline. The verified, grounded, lived interaction,, the event, has no persistent governing authority over the representation. The system is correct relative to its training data and wrong relative to the user's actual reality. Every time.

In medicine, the patient's chart, a coordinate representation assembled from lab values, diagnostic codes, procedure records, and clinical notes, becomes the governing description of the patient's medical reality. The patient's lived experience of the illness, the pain that doesn't show up on imaging, the fatigue

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that no lab value captures, the cognitive fog that no code names, the cascading consequences across family, work, identity, and daily function, is present in the room but absent from the governing record. The chart outranks the patient. Decisions are made that are correct relative to the chart and wrong relative to the event.
In law and bureaucracy, the case file, the eligibility form, the procedural checklist become the governing representation of a person's situation. The lived reality, the complexity, the context, the stakes, the power asymmetries, the consequences that no form captures, is filtered out before the system ever evaluates the case. The system can follow every procedure correctly and produce an outcome that is adequate to the file and catastrophic to the person.

These are not analogies. They are not metaphors. They are not coincidences.

The Missing Coordinate Theory proposes that they are all instances of a single ontological error: modern systems treat the measured representation of an event as if it were the event itself. This error is not a design flaw in any particular system. It is a civilizational error in ontology that propagates across every domain that inherits it.

The paper formalizes this claim, identifies the mechanisms by which the error propagates, and shows that the same formal structure, the same axioms, the same filters, the same regime of bounded correctness, applies across quantum foundations, black-hole cosmology, AI architecture, institutional governance, and the phenomenology of chronic illness.

1.2 Primitive Events vs. Coordinate Events (First Intuitive Pass)

The theory rests on one foundational distinction.

A primitive event $\mathcal{E}$ is the actual occurrence: a change in the world as lived, undergone, propagated, and consequentially real across experiential, biological, material, relational, and historical domains. An event is not merely a point in spacetime. It is what happened, to whom, how, why, under what conditions, with what stakes, with what consequences, and with what transformation across bodies, relations, environments, and history.

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Coordinates do not create an event. They only locate it.
Given the coordinates:

- November 26, 2026, 2:00 PM, 123 Main St, Elizabeth City, NC

the time and place are specified. The event is not. Who was involved, what changed, how it changed, why it mattered, and what followed remain entirely unspecified. Coordinates are sufficient to index an event but not to exhaust it.

A coordinate event $C_i(\mathcal{E})$ is the standardized representation of that occurrence within a measurement grid. It is any description of the event in the language of a particular system, observer, or apparatus:

- A spacetime point (t, x, y, z)
- A quantum state vector $|\psi\rangle_{S|O_i}$
- A medical ICD diagnostic code
- A legal charge in a case file
- A line in a database
- A KPI on a dashboard
- A log entry in an AI system
- A timestamp on a feeding chart

Each of these is a coordinate event. Each indexes something that happened. None of them exhausts what happened.

The subscript i in $C_i(\mathcal{E})$ matters. It marks the fact that every coordinate representation is relative to a particular observer, system, institution, or apparatus. There is no *"view from nowhere",* no $C_{univ}(\mathcal{E})$ that captures the event for all observers simultaneously. This observer-relativity is not a limitation to be overcome. It is a structural feature of how events relate to the systems that represent them. (This claim is formalized as Axiom 6 in Section 4.)

The Missing Coordinate Theory begins from the claim that modern systems routinely mistake the second for the first. The coordinate event, the chart, the

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file, the code, the wavefunction, the statistical prediction, is treated as if it were the primitive event. The representation acquires governing authority. The event is demoted to *"input," "evidence," "subjective report," or "noise."*

The system-subject is the person a system is about, serves, evaluates, or governs, but who is not granted primary authority to define reality in a governing form. The system-subject may be a patient, a user, a client, a customer, or any governed participant. This person is positioned closest to the primitive event $\mathcal{E}$ yet structurally least authorized to make it count. The system-subject's testimony is received but not admitted to governing status. Their reality can be named, recorded, acknowledged, and still remain powerless.

1.3 The 9/11 Example

The example that makes this distinction sharpest requires no technical background.

Consider two coordinate descriptions:

September 11, 2001, 6:00 AM, World Trade Center, New York City
September 11, 2001, 6:00 PM, World Trade Center, New York City

The coordinates register that an immense change occurred. Same location. Same date. Twelve hours apart. The dimensional framework; time, place, perfectly indexes that something happened between these two points.

The coordinates do not contain the deaths. They do not contain the smoke, the toxic air, the collapsed structures, the biological damage. They do not contain the political rupture, the trauma, the grief, the fear, the policy consequences, the wars that followed, the surveillance architectures that were built, the children who grew up without parents, the first responders who developed cancers, the historical meaning that accumulated over decades.

Dimensions can register that reality changed while remaining unable to carry what that change was.

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This is not a failure of these particular coordinates. It is a structural feature of all coordinate representations. The coordinates $C_i(\mathcal{E})$ locate the transformation. They do not contain the transformed reality. They measure the delta. They do not carry the event that changed the world.

The coordinates are not wrong. They are not useless. They are accurate and necessary. But they are not the event. They are an index of the event. And when systems treat the index as if it were the thing, when the coordinate description acquires governing authority over the primitive event, the result is a system that is correct about the coordinates and wrong about everything that matters.

This is what the theory calls ontological inversion.

1.4 Ontological Inversion

The proper ontological order is straightforward: events come first, and representations are derived from them.

$$\text{event} \rightarrow \text{measurement / representation}$$

An event occurs. Someone observes it, measures it, records it, codes it, models it. The representation is a secondary artifact, a tool created to index, compare, communicate, and act on the event. Dimensions time, space, energy, temperature, amplitude, diagnostic code, legal status, database field were invented as measurement abstractions to track differences between events in standardized, shareable form. They are excellent instruments for formal comparison. They do not, by themselves, carry the full reality of what changed.

Modernity inverted this order.

$$\text{measurement / representation} \rightarrow \text{what counts as the event}$$

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Instead of representation serving the event, we began treating dimensional, coordinate-compatible representations as if they were exhaustive of reality itself. The measurement became the authoritative substitute for the actual, lived occurrence. What can be measured, coded, standardized, and formally represented was elevated to governing status. What cannot be measured in those terms relational context, lived transformation, stakes, consequences, historical meaning, the experience of the person closest to the event was demoted to secondary status: *"subjective," "anecdotal," "soft," "not evidence-based," "outside our scope."*

This inversion did not happen by accident. It has a specific philosophical origin.

Descartes distinguished between kinds of reality: *res extensa* (extended, measurable, spatial substance) and *res cogitans* (thinking, experiential, first-person substance). This was a distinction a recognition that reality has different aspects. It was not, in Descartes' formulation, a hierarchy that required one to dominate the other. Distinguishable aspects of reality can still co-govern the same event.

But society took that distinction and institutionalized it as a hierarchy of authority. Externally measurable reality the kind that can be quantified, standardized, replicated, and formally represented was elevated to primary governing status. Lived, relational, biological, contextual, and first-person reality was demoted to secondary status. The error was not the making of distinction. The error was the institutionalization of hierarchy.

Science refined this hierarchy into admissible knowledge: what counts as evidence, what counts as a valid observation, what counts as a legitimate variable. The peer review process, the experimental method, the statistical framework all powerful, all necessary also enforce a boundary around what kinds of reality may enter the governing conversation. Reality that does not fit measurement-first, standardized form is not merely underweighted. It is rendered structurally non-governing.

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Institutions operationalized this hierarchy into procedure. The hospital chart, the legal file, the eligibility form, the risk score, the KPI dashboard each is a coordinate system designed to represent events in a standardized, actionable form. Each encodes the hierarchy: what can be formally represented governs; what cannot be formally represented does not.

Systems enforce this hierarchy as though it were neutral reality. The system does not announce that it is filtering out aspects of the event. It does not acknowledge that its coordinate representation is partial. It simply acts on what it can see and treats what it cannot see as if it does not exist.

The result is that systems do not govern events directly. They govern authorized reconstructions of events. The reconstruction is filtered through the system's inherited ontology, which determines what kinds of reality count before the system is ever applied to any particular situation.

A system can be internally valid, procedurally correct, and empirically supported while still governing an inadequate reconstruction of the event it claims to address.

This is what the theory calls ontological inversion: the coordinate representation $C_i(\mathcal{E})$ has acquired governing authority over the primitive event $\mathcal{E}$ it was created to serve. Formally: $\text{Inversion}(i) := C_i(\mathcal{E}) \succ \mathcal{E}$
where "$\succ$" means *"is treated as having governing authority over."*

The inversion is historical, social, and institutionalized not merely logical. It is inherited by every system built on the assumption that measurable representation is exhaustive of the event. It is not a conspiracy. It is not malice. It is architecture.

1.5 Why the Theory Says *Incomplete*, Not *Incorrect*

The Missing Coordinate Theory is not anti-science. It is not anti-measurement. It is not anti-formalization.

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It does not claim:

- that science is false or that scientific method is illegitimate;
- that measurement is useless or coordinates are meaningless;
- that institutions are always malicious or always self-consciously exclusionary;
- that first-person or subjective report should automatically override all formal process;
- that quantitative methods should be replaced by qualitative methods;
- that the coordinate representation is wrong.

The theory claims something more precise.

Scientific measurability is powerful but incomplete as a total ontology for events.

The problem begins not with measurement itself but with the civilizational decision to treat one translation of an event as the whole event. Measurement captures real aspects of real events. It does so with extraordinary precision, reliability, and generalizability. These are genuine virtues. The theory does not deny them.

What measurement does not do what no coordinate system does is exhaust the event. The coordinate representation $C_i(\mathcal{E})$ captures aspects of $\mathcal{E}$ that fit into the coordinate space M_i. It necessarily excludes aspects that do not fit. This exclusion is not a bug. It is a structural consequence of the fact that any finite-dimensional representation projected from a higher-dimensional object is lossy. (This is formalized in Axiom 2: Non-Exhaustibility of Representation, Section 4.)

The correction implied by the theory is therefore not the abandonment of measurement. It is the restoration of measurement to its proper place:

- The event $\mathcal{E}$ is primary. Dimensional description $C_i(\mathcal{E})$ is one translation of the event, not its exhaustive account.

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- Measurable structure is real and necessary, but not sufficient as a complete ontology for events.
- Lived, relational, biological, and contextual structure are also constitutive of events.
- No system should treat formal representation as the whole event.
- The person closest to the event the system-subject must not be structurally disqualified from helping define it in governing form.
- Authorized reconstruction must remain accountable to the event it claims to represent.

The real correction is not a technical revision. It is a restoration of governing authority.

Representation must serve the event, not replace it.

1.6 What the Theory Is NOT

Because the theory diagnoses a structural error in how modern systems handle reality, it is vulnerable to predictable misreadings. This section names them explicitly so the reader does not have to guess.

The theory is not anti-science. It depends on science. Axiom 6 (Non-Universal Measurability) is grounded in general relativity's observer-dependent proper time and Relational Quantum Mechanics' rejection of absolute quantum states. The formal operators in Section 6 use information theory, fiber-bundle geometry, and KL divergence. The theory is a contribution to scientific ontology, not a rejection of it.

The theory is not anti-measurement. It requires measurement. Coordinate representations $C_i(\mathcal{E})$ are real, necessary, and powerful. The theory's claim is that they are incomplete, not that they are wrong. A map that covers 80% of the terrain is enormously useful. Mistaking it for the terrain is the error.

The theory is not anti-formalization. The entire paper is a formal construction: axioms, propositions, definitions, operators, functionals, worked proofs, and geometric structures. The theory uses formal tools to diagnose the

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limits of formal tools. This is not a contradiction. It is the normal business of mathematics and logic.

The theory does not claim that subjective report automatically overrides formal process. The system-subject's testimony is not automatically correct. It is, however, automatically relevant because the system-subject is positioned closest to the primitive event. The theory does not propose replacing institutional governance with personal narrative. It proposes restoring the event to governing status, which includes restoring the system-subject's capacity to introduce reality in a form that can challenge, correct, and enrich the coordinate representation.

The theory is not relativism. It does not claim that all descriptions are equally valid or that truth is merely perspectival. It claims that event reality is observer-relative (different observers have access to different aspects of the same event) and that no single observer's coordinate system exhausts the event. This is not *"anything goes."* It is *"no single coordinate system captures everything,"* which is a precise structural claim with formal consequences the same claim that general relativity makes about simultaneity and that **Relational Quantum Mechanics** makes about quantum states.

The theory is not a demand to feel more. It is not an argument for empathy over rigor, for emotion over logic, for the heart over the head. It is an argument that the dimensional frameworks we use to govern events are structurally incomplete, and that the missing dimensions lived transformation, relational context, stakes, consequences, persistent corrected memory are real aspects of events that must be explicitly modeled if systems are to govern events rather than their own authorized projections.

1.7 Scope and Structure of the Document

The paper develops the Missing Coordinate Theory in four movements: formal foundation, domain application, empirical grounding, and self-diagnosis.

Formal foundation (Sections 2 - 6).

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Section 2 establishes the full formal vocabulary and notation, defining every symbol, its type, and its role in the theory. Section 3 defines the foundational objects: primitive events, coordinates, and ontological inversion. Section 4 presents the axiomatic framework: Axioms 1 - 5 establish event priority, non-exhaustibility, the inversion, the Cartesian hierarchy, and systemic inheritance.
Axiom 6: Non-Universal Measurability is the theory's technical heart, establishing that no single coordinate chart can exhaustively capture all primitive events for all observers, with direct connections to GR and RQM. Corollary 6.1 provides a proof-by-contradiction using memory divergence as empirical evidence. Section 5 derives eight propositions: the governing ontology, the four filters (ontological, authority, interpretive, correctness), the Regime of Bounded Correctness, the Named Reality Problem, cross-domain recurrence, and Proposition 8

The Missing Dimensions which provides the constructive complement to Axiom 6 by identifying what the coordinate representation excludes. Section 6 introduces the formal operators and quantification: Definition 1 (the relational event definition that addresses an identified gap in RQM), Definition 2 (the interaction operator for cross-perspective alignment), the relational actualization condition, the persistent correction operator for machine cognition, the information-theoretic extension, the governing-authority loss functional, the Missing-Coordinate Theorem, and the fiber-bundle formulation.

Domain application (Sections 7 - 12).
Section 7 applies the framework to physics: GR as precedent for non-universal measurability, the quantum measurement problem as coordinate inversion, Relational Quantum Mechanics (including the Faglia gap and the MCT solution via Definition 1), QBism as a partial neighboring move, LQG and background independence, black-hole cosmology, Einstein Cartan torsion bounce, supermassive black holes as stabilizers, and the JWST/JADES anomaly. Section 8 develops the relational model of shared reality: how objectivity emerges through interaction rather than existing as a pre-given global object, with everyday examples (dinner, restaurant choice) and scientific examples (Wigner-style divergence, horizon case). Section 9 presents three fully worked

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examples using a consistent 7-step template: Wigner's Friend (full three-phase walkthrough with summary table), Dinner/Taco Tuesday, and Black-Hole Horizon. Section 10 formalizes the machine cognition application: reasoning as memory applied forward (Damasio, Friston), architectural amnesia in current AI, the inversion in AI, the missing coordinate proposition, the persistent correction operator in full, the stateless-vs-reasoning comparison table, and architectural implications. Section 11 presents the four filters and bounded correctness as a unified institutional mechanism, including the closed admissibility loop, the loss functional in mechanism form, and the Named Reality Problem with applied examples. Section 12 applies the filters to specific domains: medicine, law and bureaucracy, governance and metrics, academia and knowledge production, with the system-subject and authority asymmetry made explicit.

Empirical grounding (Sections 13 & 14).
Section 13 presents phenomenological and empirical case studies, centering the essay **Living Without Autopilot** as evidence of the theory's core claims in lived form: the collapse of predictive compression as event decomposition, the brutal economy of pain as missing dimensions, medical fragmentation as ontological inversion, the feedback loop as closed-loop instability, dependency as relational catastrophe, and the construction of manual survival architecture as a human analogue of persistent corrected memory. Section 14 deploys the scenario dataset across seven additional domains: coordination and expectation, measurement and observation, logging vs. lived reality, catastrophe and propagation, identity and category pressure, continuity and memory loss, and stateless vs. stateful AI each mapped through the 7-step scenario template.

Self-diagnosis and completion (Sections 15 - 21).
Section 15 provides the quantification and geometric upgrade: the loss functional in full, the fiber-bundle structure of missing dimensions, and the bridge to physics. Section 16 synthesizes the unified interpretation. Section 17 states the final formal statement of the theory in canonical four-point form plus quotable paragraph. Section 18 applies the theory to itself: the paper as coordinate representation, the theory's creation as primitive event, predicted

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reception through the four filters, and the refusal to let the paper become the event. Section 19 identifies implications and future work across physics, AI, institutions, medicine, and philosophy. Section 20 provides references. Section 21 contains appendices: full axiom set, full propositions, expanded RQM formalism, Einstein/Cartan technical notes, machine cognition supplement, equations as indexing tools, plain-language summary, and the full text of "Living Without Autopilot."

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